



A comprehensive study on parametric optimization of the pin-fin heat sink to improve its thermal and hydraulic characteristics

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ABSTRACT

In an effort to reduce the highest temperature of electronic pieces, using high performance cooling systems is required, and for this reason, optimization is one of the most essential and attractive issues that has an undeniable role in industry and technology. In this study, a finite volume method is employed to numerically simulate conjugate heat transfer in pin-fin heat sinks, which are of the most common cooling systems, and all the geometric parameters affecting their performance are optimized based on thermal and hydraulic performances (FOM). These geometric parameters are: the number of fins, the fin height, the fin diameter, and the transverse pitch. It will be shown that the total performance of the pin-fin heat sink improves step by step by optimizing these parameters. Also, it will be suggested that the shape of pin-fins changes to a tapered one. It is proved that changing the fin shape has a significant influence on the convective heat transfer coefficient and pressure drop and enhances the total performance of the system. Furthermore, it is observed that the total performance of the pin-fin heat sink increases more than 17% during the stages of the optimization.

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1. Introduction

During the previous decades, advances in technology have led to the production of low-volume electronic components with high processing capabilities. One of the most critical concerns about these systems is the excessive rise in temperature that causes damage to these components. So, the main challenge of these systems is controlling their highest temperature by utilizing cooling systems. Nowadays, the requirement of effective and high-performance cooling systems is undeniable. The most common cooling techniques utilized in the field are fins [1], heat pipes [2], and microchannel heat sinks [3].

Among all methods, because of being convenient and low-cost, cooling by passing the air flow through finned microchannels is a popular approach [4]. The fins placed on finned microchannels are generally divided into two types: plate- and pin-fins. Many investigations have already been performed to study the plate-fin heat sinks [5–7]. Nilpueng et al. [8] experimentally explored the effect of different parameters such as the velocity of the fluid, the temperature of the heat sink base and phase shift on the thermal and hydraulic performance of sinusoidal wavy plate-fin heat sinks (SW-

PFHSs) and cross-cut sinusoidal wavy plate-fin heat sinks (CCSW-PFHSs). They observed that increased phase shift and air velocity contribute to heat transfer and pressure drop enhancement while the heat sink base temperature exhibits a subtle impact on the hydrothermal performance. Additionally, CCSW-PFHSs demonstrate a higher Nusselt number than that of SW-PFHSs by about 5.9–19.1% with the same phase shift. Tariq et al. [9] proposed two novel plate-fin heat sinks (with perforations and slots) and analyzed them both experimentally and numerically. While exploiting the advantage of decreased weight due to slots and perforations, they observed higher heat transfer coefficient and lower pressure drop in novel plate-fin heat sinks (NPFHSs) compared to those of conventional plate-fin heat sinks.

Pin-fin heat sinks have also been explored extensively in the literature [10–12]. Wang et al. [13] numerically studied a novel configuration of micro latticed pin-fins, mixing the ordinary pin-fins and lattice structure. Their results showed that utilizing the lattice pin-fin causes an enhancement in total pressure drop in comparison with the conventional pin-fin. Soleymani et al. [14] numerically studied a hybrid pin-fin microchannel heat sink and investigated the impact of various parameters such as the configuration and angle of pin-fins, fluid velocity and heat flux on the thermal performance of the system. They found that rectangular pin-fins with rounded edges have a better effect on the thermal performance in

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Nomenclature

A	Area (m^2)
C_p	Air thermal capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
d	Pin-fin diameter (mm)
h	Convective heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
H	Pin-fin height (mm)
k	Thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
l	Heat sink length (mm)
N	Number of pin-fins
P	Pressure (Pa)
Q	Total heat (W)
r	Tapered pin-fin tip diameter (mm)
R	Thermal resistance (K W^{-1})
s	The axial pitch
t	The transverse pitch
T	Local temperature (K)
$\mathbf{U}$	Velocity vector (m s^{-1})
W	Heat sink width (mm)

Subscripts

a	Air
f	Fluid
max	Maximum
min	Minimum
opt	Optimum
s	solid

Greek

δ	The heat sink thickness (mm)
η	Fin efficiency (-)
μ	Viscosity (Pa s)
ρ	Density (kg m^{-3})

Superscripts

*	Non-dimensionanl
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comparison with NACA airfoils. Li et al. [15] numerically examined a novel microchannel heat sink with isosceles triangular cavities and various pin-fins, including rectangular, streamlined, backward drop-shaped, and forward drop-shaped fins for different Reynolds number and flow rates. Zeng et al. [16] suggested a new configuration for microchannel heat sinks based on topology-optimized heat sink structures, comprising two types of fins, oblique and trapezoidal fins, arranged in a semi-wavy pattern. The growth of thick boundary layers is prevented due to the non-continuous fin design. By comparing the new design with five conventional fin patterns, they observed that the novel design shows the best performance in the corresponding flow range.

Recently, researchers have also focused their attention on novel heat sink configurations such as double-layer microchannel heat sinks (DMHS), which were originally proposed by Vafai and Zhu [17], after whom other researchers such as Saidi and Khiabani [18] investigated the effect of increasing the number of layers. Xie et al. [19] found that not only are DMHSs capable of improving the thermal performance of heat sinks, but they also cause a lower pressure drop in the channels. There are also some efforts on optimizing the geometric parameters of DMHSs in the literature [20–22]. In a recent study by Zhang et al. [23], the authors studied a modified structure of DMHSs with an upper layer comprising dense, narrow fins while the lower layer has thick fins with a relatively high distance from each other. They showed that this configuration reduces the thermal resistance of the system by 9.42% compared with traditional DMHSs with the same pumping power.

Additionally, vortex generators are shown to be an effective way of enhancing heat transfer in heat sinks while they do not raise

the pressure drop significantly due to inducing a swirling turbulent flow in the gap between fins [24]. Lu and Zhai [25] studied microchannel heat sinks with both dimples and vortex generators and showed an increase in both thermal performance and pressure drop. They also found that the “common flow up” configuration of vortex generators shows better heat transfer performance in comparison with the “common flow down” configuration and proposed the optimal values for the height ratio and attack angle of vortex generators. Recently, Bahuguna et al. [26] employed an entropy generation method to evaluate the hydrothermal performance of heat exchangers with triple blade vortex generator inserts and obtained a geometry with the highest Nusselt number.

Modifying the shape of pin-fins is another applicable way of enhancing the thermal performance of heat sinks [27]. Turkiyilmazoglu conducted valuable studies to obtain exact formulas for the heat transfer characteristics of growing exponential- and parabola-shaped fins and compared their performance with that of plain pin-fins [28,29]. In order to induce vortical flows, Chang et al. [30] explored the hydrothermal performance of microchannel heat sinks with twisted-tape pin-fin arrays. They reported thermal performance factors (TPF) in the range of 1.55–1.39, showing a better performance in comparison with other conventional pin-fin arrays in the literature. They also proposed two empirical correlations for calculating the Nusselt number.

One of the newest techniques for increasing the efficiency of pin-fin heat sink is using water [31] and nanofluids [32] instead of air as the cooling fluid. Khoshvaght-Aliabadi et al. used Al_2O_3 /water nanofluid as the cooling fluid in pin-fin miniature heat sink and compared it with water. They proved that utilizing the nanofluid has a remarkable influence on the convective heat transfer coefficient and it could improve the thermal performance up to 15%.

Nowadays, in search of achieving better performance and increasing the performance of systems, optimization has received a lot of attention. Therefore, many studies have been done on the optimization of pin-fin heat sinks. Yang et al. [33] tried to optimize the geometrical properties of pin-fin heat sinks experimentally. They could increase the total thermal performance of the pin-fin heat sink more than 10%. Zhao et al. [34] performed a structural optimization method on micro square pin-fin heat sinks by changing the located angle and porosity of pin-fins. Their results revealed that the maximum thermal performance of the heat sink was reached while the located angle and porosity were 30° and 0.75, respectively.

Heat transfer enhancement is usually accompanied by increased pumping power, and since an increase in pumping power is not economically justified, heat transfer improvement and pumping power increase should be considered simultaneously so that an optimal case is obtained. As a result, optimizing heat sinks for achieving the best thermal performance with the least increase in pumping power is of utmost importance. Hong and Cheng [35] performed a numerical investigation on optimizing offset strip-fin microchannel network heat sinks. They considered the influence of geometrical properties on the fluid flow and found an optimum strip-fin size that minimizes the pumping power, which depends on the input heat flux and maximum wall temperature. Moreover, the optimum design for the heat sinks with micro square pin-fins performs better than the heat sinks with column pin-fins in terms of thermal performance. Ghahremannezhad and Vafai [36] performed an extensive analysis to find an optimum design for finned heat sinks with porous substrates. Given their results, there is an optimal porous design for each set of the heat sink geometrical parameters. Testing a wide range of Reynolds numbers and heat sink material and porosities, they observed the dominance of porous heat sinks over ordinary ones. John et al. [37] utilized a figure of merit (FOM) criterion to evaluate the hydrothermal performance of both square-shaped and circular pin-fin heat sinks

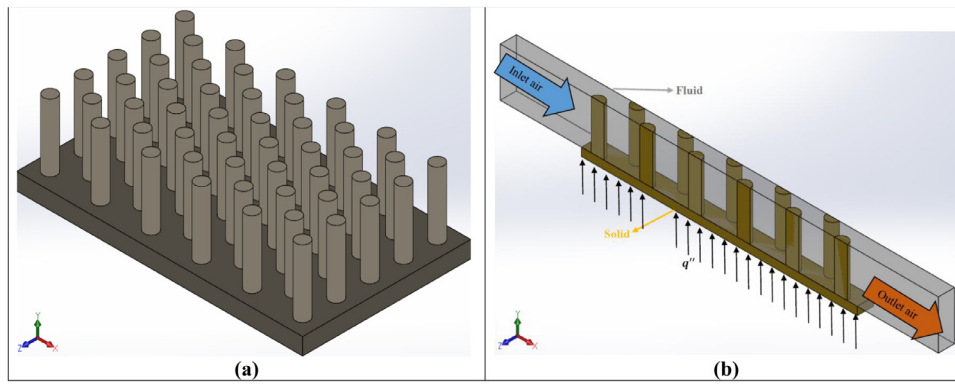


Fig. 1. The 3D schematic of (a) the pin-fin heat sink and (b) computational domain.

for various geometrical parameters. Considering limited ranges for them, they studied the effect of each of the parameters on FOM. At high Reynolds numbers (300–500), the square pin-fin heat sinks decreased the highest temperature of the system more than those with circular pin-fins while for low Reynolds numbers (50–300), the opposite trend is observed.

To give a practical guideline for designing high performance pin-fin heat sinks with the best hydrothermal performance, in this study, a comprehensive geometrical optimization of a conventional pin-fin heat sink is performed. The geometrical parameters such as the number fins (N), fin diameter (d), longitudinal pitch (s), fin height (H) and transverse pitch (t) will be considered in this study, and it will be tried to enhance the thermal and hydraulic performance of the pin-fin heat sink by optimizing these parameters. In order to take both hydraulic and thermal performance of heat sinks into account, the figure of merit (FOM) will be employed as the comparison criterion. The optimization will be carried out in five stages and all the parameters will be optimized step by step. As the last step of optimization, a novel design of fins will be investigated. In this design, the shape of pin-fins changes to tapered ones (the diameter of fins is kept non-uniform and changes from the base to tip), to study the effect of tapered pin-fins on FOM, and the optimum value for the tip diameter will be obtained. Finally, the influence of the Reynolds number (Re) on the thermal and hydraulic performance of the system will be investigated and reported.

2. Problem statement

2.1. Description of the physical system

Figure 1 (a) indicates a 3D schematic of a heat sink with circular pin-fins arranged in the staggered configuration. The pin-fin heat sink is exposed to a uniform heat flux from the bottom surface and air passes through the finned channels to reduce the temperature of the system. Because of the symmetry geometry, only a part of it with symmetry boundary conditions, shown in Fig. 1 (b), is simulated. Pin-fins are arranged in a staggered configuration and all the properties and characteristics of the fins such as the material, height and diameter are the same. According to Fig. 1 (b), the heat sink bottom wall is subject to a constant heat flux, and air is used as the working fluid for cooling the heat sink. Fig. 2 (a) and (b) demonstrate the side and top views of the pin-finned heat sink with the geometrical parameters. In this figure δ , l and w , which are constant values in all simulations, denote the heat sink thickness, length and width, respectively. Additionally, N , H , d , s and t are the geometrical parameters, which vary and are supposed to be optimized. These parameters represent the number of fins, the fin height, the pin-fin diameter, longitudinal pitch and transverse

Table 1

Geometrical and physical parameters and constants.

	Parameters	Lower value	Constant value	Upper value
Geometrical	d (mm)	0.7	–	2
	H (mm)	7.5	–	20
	l (mm)	–	100	–
	r (mm)	0.17	–	0.85
	s (mm)	4	–	8
	t (mm)	1.66	–	49
	δ (mm)	–	2.5	–
Physical	ρ_{fluid} [$kg\,m^{-3}$]	–	1.225	–
	ρ_{solid} [$kg\,m^{-3}$]	–	2719	–
	$C_{p,fluid}$ [$J\,kg^{-1}\,K^{-1}$]	–	1006.43	–
	$C_{p,solid}$ [$J\,kg^{-1}\,K^{-1}$]	–	871	–
	k_{fluid} [$W\,m^{-1}\,K^{-1}$]	–	0.0242	–
	k_{solid} [$W\,m^{-1}\,K^{-1}$]	–	175	–
	μ [$kg\,m^{-1}\,s^{-1}$]	–	0.00001789	–

pitch, respectively. For summary, the values of all geometric and physical parameters are given in Table 1. It is worth mentioning that for obtaining valid, physical results, the air domain is started 50 mm before the heat sink and extended 50 mm after the finned area.

2.2. Mathematical model and solution

The conjugate method is utilized to simulate heat transfer in both fluid and solid domains through the finite volume method. The air flow with constant thermophysical properties is considered to be three-dimensional, steady-state, and incompressible. Since Reynolds numbers are higher than 2700, which falls into the turbulent regime, turbulent three-dimensional Navier-Stokes equations combined with continuity and energy equations are solved numerically. Continuity, Reynolds Average Navier-Stokes, and energy equations are as follows [38]:

$$\nabla \cdot \mathbf{U} = 0 \quad (1)$$

$$\rho \mathbf{U} \cdot \nabla \mathbf{U} + \nabla \cdot (\overline{\rho \mathbf{u}' \otimes \mathbf{u}'}) = -\nabla P + \nabla \cdot \mu (\nabla \mathbf{U} + (\nabla \mathbf{U})^T) \quad (2)$$

$$\rho C_p \mathbf{U} \cdot (\nabla T_f) = k_f \nabla^2 T_f \quad (\text{for fluid}) \quad (3)$$

$$k_s \nabla^2 T_s = 0 \quad (\text{for solid}) \quad (4)$$

$k - \varepsilon$ along with eddy viscosity methods are used to account for turbulent effects. The eddy viscosity closure is as follows:

$$\rho (\overline{\mathbf{u}' \otimes \mathbf{u}'}) = \frac{2}{3} \rho k - \mu_t (\nabla \mathbf{U} + (\nabla \mathbf{U})^T) \quad (5)$$

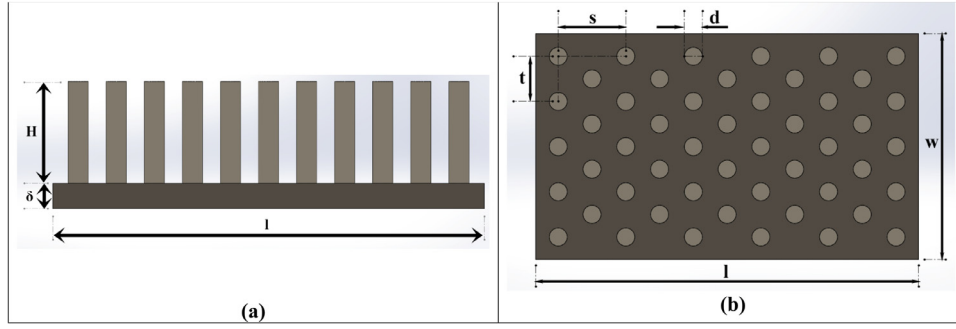


Fig. 2. The (a) side and (b) top views of the heat sink with geometrical parameters.

Transport equations for k and ε are as follows:

$$-\nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right] + \rho \mathbf{U} \cdot \nabla k = \frac{1}{2} \mu_t (\nabla \mathbf{U} + (\nabla \mathbf{U})^T)^2 - \rho \varepsilon \quad (6)$$

$$\begin{aligned} &-\nabla \cdot \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] + \rho \mathbf{U} \cdot \nabla \varepsilon \\ &= \frac{1}{2} C_{\varepsilon 1} \frac{\varepsilon}{k} \mu_t (\nabla \mathbf{U} + (\nabla \mathbf{U})^T)^2 - \rho C_{\varepsilon 2} \frac{\varepsilon^2}{k} \end{aligned} \quad (7)$$

where

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (8)$$

Empirical constants are chosen as follows: $C_{\varepsilon 1} = 1.44$, $C_{\varepsilon 2} = 1.44$, $C_\mu = 1.0$, and $\sigma_\varepsilon = 1.3$ since these values account for the turbulence well in the Reynolds range explored in this study [38,39].

Boundary conditions for solving the flow are as follows: no-slip boundary condition for all the walls adjacent to the fluid, adiabatic boundary condition for the channel top wall, fin tip, channel non-finned bottom walls, and all heat sink walls except its bottom and top walls. The heat sink bottom wall is subject to a constant heat flux of 100 W/cm^2 . At the inlet, air enters the domain with a constant velocity, whose inlet temperature is 298 K . Pressure outlet boundary condition is set for the outlet boundary. Finally, the lateral walls of the computational domain are symmetry.

To take both thermal and hydraulic performances of the heat sinks into account, a criterion, which is named figure of merit (FOM), is defined as follows:

$$FOM = \frac{\frac{h_{\text{finned}}}{h_{\text{no-fin}}}}{\left(\frac{\Delta P_{\text{finned}}}{\Delta P_{\text{no-fin}}} \right)^{1/3}} \quad (9)$$

where h and ΔP are the convective heat transfer coefficient and pressure drop, respectively. An approach based on thermal resistances is utilized for calculating the convective heat transfer coefficient. The total resistance between the heat sink bottom wall and the air flowing through fins comprises a conductive resistance in the heat sink body and a convective one in air [40].

$$R_{\text{total}} = R_{\text{conductive}} + R_{\text{convective}} \quad (10)$$

$R_{\text{conductive}}$ And $R_{\text{convective}}$ are defined as follows:

$$R_{\text{conductive}} = \frac{\delta}{k_s \cdot \frac{s}{2} \cdot l} \quad (11)$$

$$R_{\text{convective}} = \frac{1}{h \cdot A_{\text{effective}}} \quad (12)$$

where

$$A_{\text{effective}} = \frac{s}{2} \cdot l - \left(\frac{n}{2} \cdot \pi \cdot \frac{d^2}{4} \right) + \left(\frac{n}{2} \cdot \eta \cdot \pi \cdot d \cdot H \right) \quad (13)$$

Table 2
Grid independence verification.

Number of elements	$\Delta P (\text{Pa})$	$\frac{ \Delta P^i - \Delta P^{i-1} }{\Delta P^{i-1}} \times 100$	$T_{\text{avg}} (\text{K})$	$\frac{ T_{\text{avg}}^i - T_{\text{avg}}^{i-1} }{T_{\text{avg}}^{i-1}} \times 100$
234983	42.26	–	308.19	–
403864	45.621	7.95	308.0036	0.06
742389	47.423	3.95	307.965	0.013
962691	48.301	1.85	307.9287	0.012
1207348	48.335	0.07	307.928	0.00023

In equation (13), η is the fin efficiency, defined as follows:

$$\eta = \frac{\tanh(mH)}{mH} \quad (14)$$

where

$$m = \sqrt{\frac{4h}{k_s d}} \quad (15)$$

The total thermal resistance could be calculated by the following equation too:

$$R_{\text{total}} = \frac{T_{s,\text{average}} - T_{a,\text{average}}}{Q} \quad (16)$$

where $T_{s,\text{average}}$ and $T_{a,\text{average}}$ are the average heat sink and air temperatures and Q is the total heat applied to the heat sink bottom wall. An iterative method is required for solving equations (10) to (16) to find the convective heat transfer coefficient.

As shown in Fig. 3, fluid and solid domains are meshed with quadrilateral elements, which are fined near the wall for capturing relatively high gradients. A mesh independence study is performed for finding the appropriate number of elements, and the results are shown in Table 2. According to Table 2, using the grid with 962691 elements guarantees that hydrothermal results are independent of the grid resolution.

In order to validate the numerical simulation, the simulation of a pin-fin heat sink studied by Diani et al. [38] is done. They first studied the hydrothermal performance of a plate-fin heat sink experimentally, with which they validated their numerical procedure. Then, they employed the validated numerical method to simulate pin-fin heat sinks cooled by the turbulent flow of air and investigated the effect of such parameters as the number of fins, as well as transvers and longitudinal pitches on the hydrothermal performance of heat sinks. They studied microchannel pin-fin heat sinks with different geometric parameters and performed a comparison between various cases. Fig. 4 shows a fine agreement between the obtained and available data. The maximum deviation between the two sets of data is 0.86% for α and 1.27% for ΔP . Given the small values for the relative deviation, it can be assumed that the difference observed between our study and Diani's can be attributed to the meshing method and the numerical solution errors, which

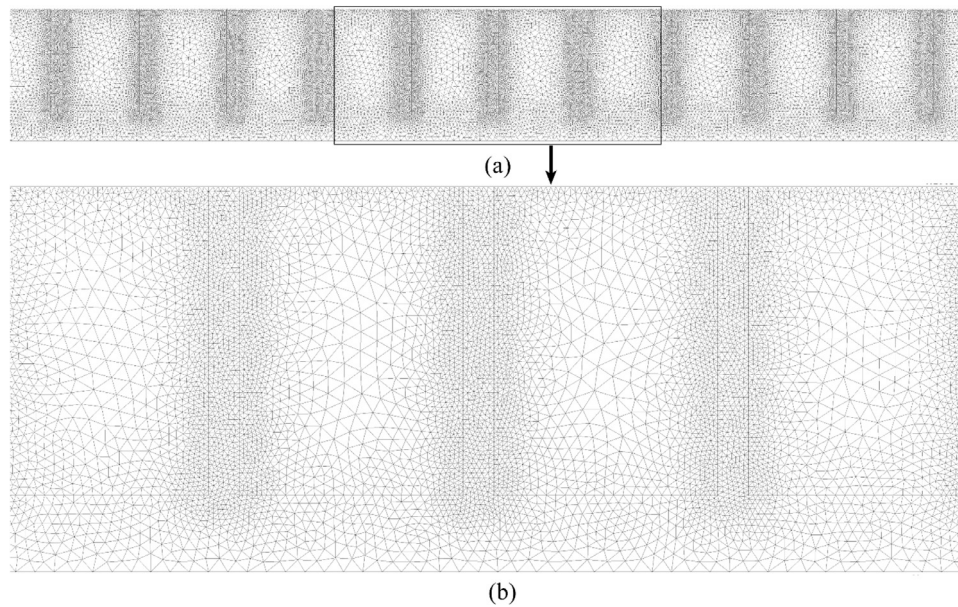


Fig. 3. The domain meshing scheme for (a) zoomed out and (b) zoomed in views.

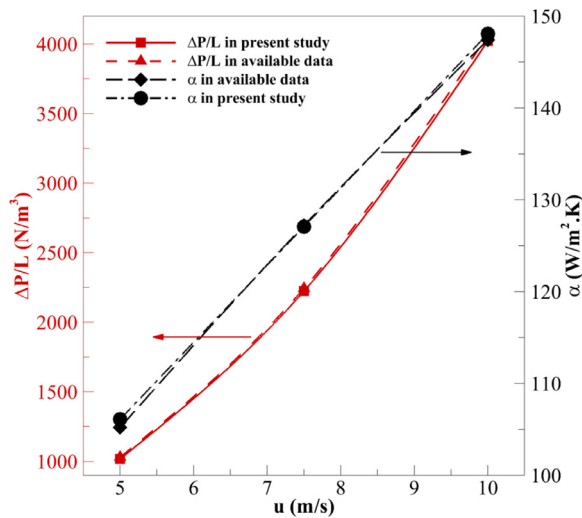


Fig. 4. The comparison of pressure drop and heat transfer coefficient between Diani's [41] study and the present work.

is negligible as the general trend is compatible with the literature data.

3. Results and discussion

To achieve the maximum performance of the pin-fin heat sink, the optimization of its geometric parameters is performed in several stages and it has been tried to improve the performance of system step by step by optimizing each parameter in each stage. The first stage of the optimization is allocated to finding an acceptable range of the fin numbers (N). To do so, for certain values of the fin diameter ($\frac{d}{l} = 0.005, 0.01, 0.015, \text{ and } 0.02$), all possible values for the number of fins (3 to 61) arranged in a staggered configuration have been investigated. For all cases, there is always one pin-fin at the beginning and one at the end of the heat sink substrate. Thus, the new pin-fins are added between these two fixed pin-fins, changing the longitudinal pitch (s). Fig. 5 (a and b) show the pressure drop and convective heat transfer coefficient for different

cases. As reported in other studies too [34,37,41,42], it can be seen from Fig. 5 that both the pressure drop and convective heat transfer coefficient increase when the number of pin-fins and their diameter increase. Since increasing the convective heat transfer coefficient and pressure drop improves the thermal performance and lowers the hydraulic performance, respectively, the FOM criterion should be used to compare them and evaluate the total performance of the pin-fin heat sink. FOM is used to find the range of the optimal number of fins, which is shown in Fig. 6. It reveals that for each value of $\frac{d}{l}$, there is an optimum value for the number of fins. It also can be seen that, as the number of fins increases, the highest value of FOM increases firstly and then decreases. In fact, there should be an optimum value for $\frac{d}{l}$ for a certain number of fins. However, these two parameters (N and d) are not independent, so they should be considered simultaneously. Considering Fig. 6, a range of fin numbers from 15 to 30 is assumed to be a probable range for which the best performance could be achieved. To verify this hypothesis, the second phase of the optimization has been carried out.

In the second stage, for the range of fin numbers specified in the first stage, simulations have been done for the cases whose $\frac{d}{l}$ changes from 0.005 to 0.02, and the results are expressed in Fig. 7. This figure shows that the increase in the pin-fin diameter augments both pressure drop and convective heat transfer coefficient. Moreover, Fig. 8 shows that for the studied range of fin numbers and diameter, the highest value of FOM corresponds to the one with $N = 26$ and $\frac{d}{l} = 0.0085$. It is worth mentioning that for having a clear vision of Figs. 7 and 8, only the 23–28 range for the number of pin-fins is depicted, as the values out of this range are far from the optimum value. Consequently, the pin-fin heat sink with $N = 26$ and $\frac{d}{l} = 0.0085$, whose FOM is 1.295, is introduced as the best configuration so far.

In the next step, the third phase, optimization has been done for the transverse pitch (t), while $N = 26$ and $\frac{d}{l} = 0.0085$. According to Fig. 9, unlike the previous parameters, both pressure drop and convective heat transfer coefficient decrease by increasing $\frac{t}{l}$, a trend also observed in other studies [34,37,41]. Also, it can be concluded that pressure drop and heat transfer coefficient have a significant growth in comparison with Fig. 9 and Fig. 7. Furthermore, it indicates that this geometric parameter (transverse pitch (t)) has an optimal value. The system experiences its maximum hydrother-

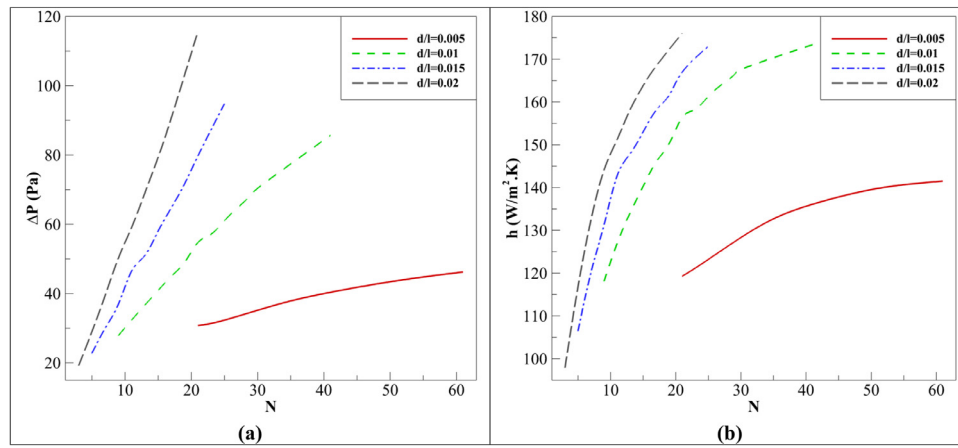


Fig. 5. The first stage of optimization results for (a) pressure drop and (b) convective heat transfer coefficient.

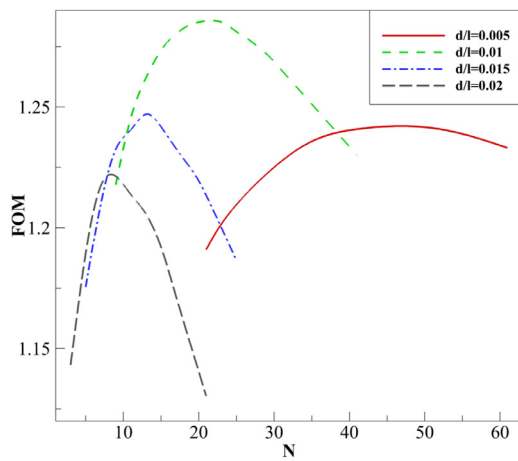


Fig. 6. The first stage of optimization results for FOM.

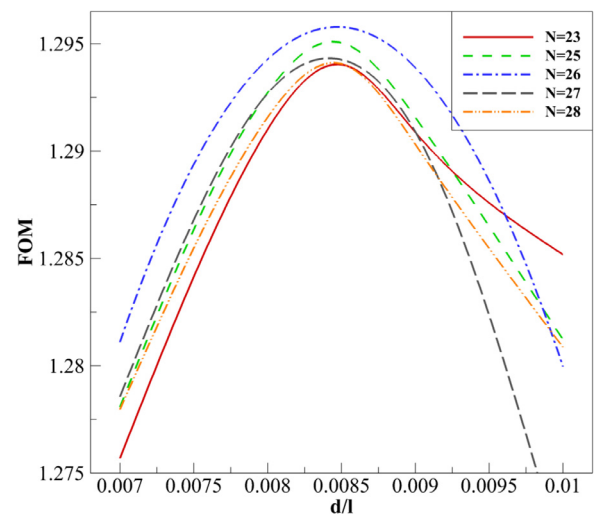


Fig. 8. The second stage of optimization results for FOM.

mal performance with an FOM of 1.334 when $\frac{t}{l} = 0.048$. This optimal value of FOM has increased 3.14% compared to the previous stage.

The fourth parameter to be optimized is the fin height ($\frac{H}{l}$), while all the previous parameters are fixed at their optimal value ($N = 26$, $\frac{d}{l} = 0.0085$ and $\frac{t}{l} = 0.048$). In this step, all possible values

for $\frac{H}{l}$ from 0.075 to 0.2 have been analyzed. As shown in Fig. 10, by increasing the fin height (H), both pressure drop and convective heat transfer coefficient decrease, which is similar to other studies in the literature [43,44]. Furthermore, like the previous stages,

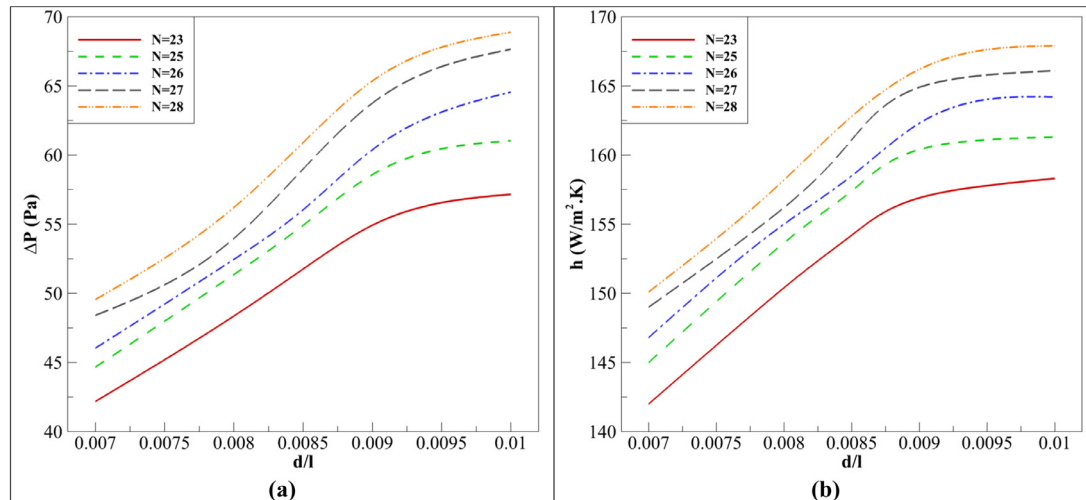


Fig. 7. The second stage of optimization results for (a) pressure drop and (b) convective heat transfer coefficient.

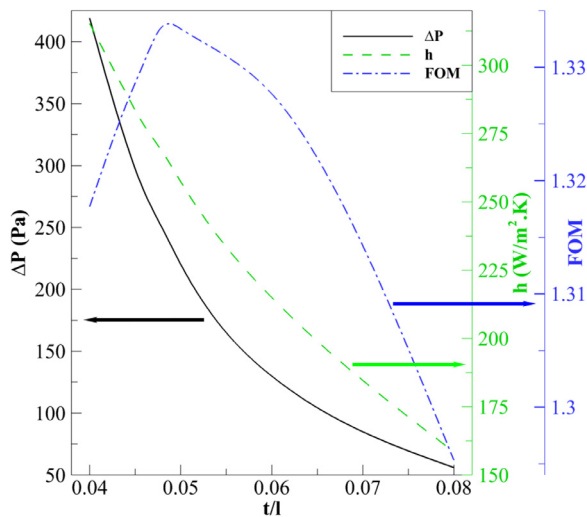


Fig. 9. The effect of (t/l) on pressure drop, convective heat transfer coefficient and FOM.

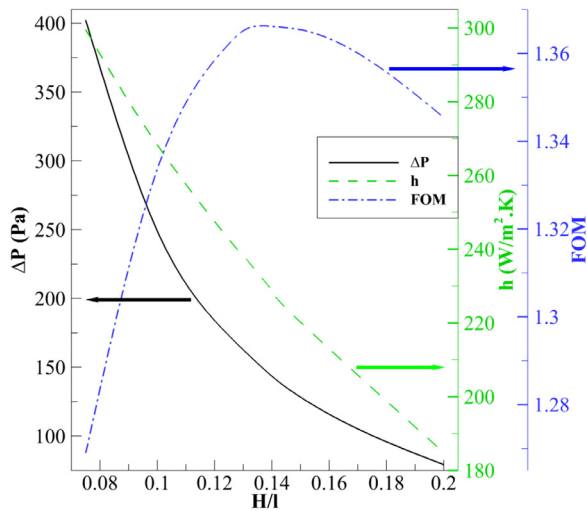


Fig. 10. The effect of (H/l) on pressure drop, convective heat transfer coefficient and FOM.

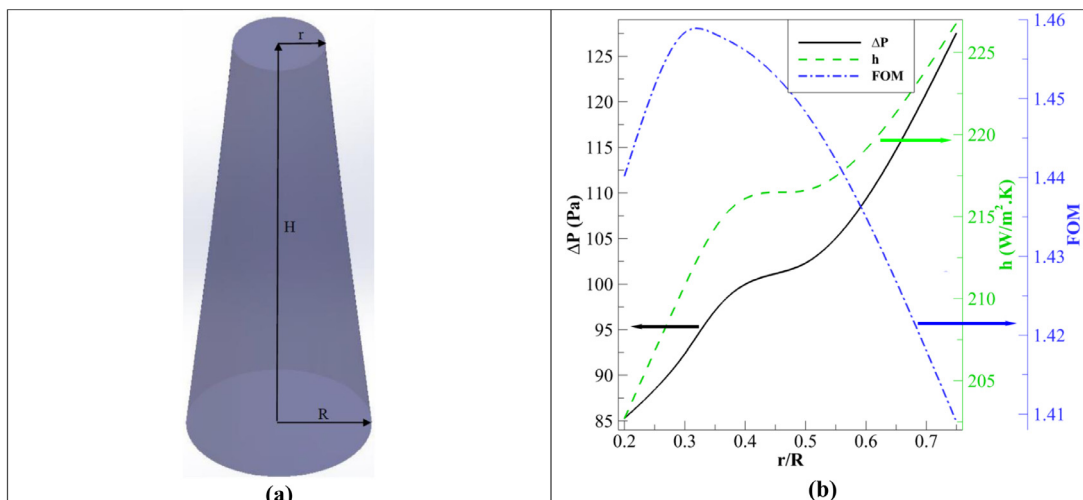


Fig. 11. a) The 3D schematic of a tapered pin-fin. b) The effect of (r/R) in tapered pin-fin heat sinks on pressure drop, convective heat transfer coefficient and FOM.

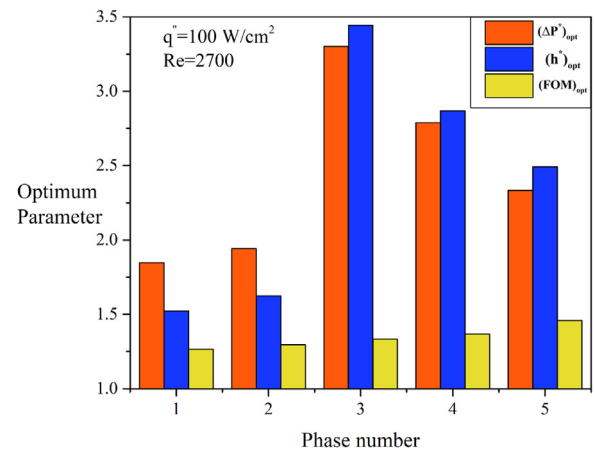


Fig. 12. Comparison of hydrothermal parameters in the optimal value of each parameters in all steps of optimization.

there is an optimal value for the fin height, at which the system has the best total performance. According to Fig. 10, the FOM analysis demonstrates that the total performance of the system increases 5% more than the previous stage when $\frac{H}{l} = 0.0135$. As mentioned, there are quite a few studies in the literature exploring the effect of geometric parameters on the performance of heat sinks. Nevertheless, in almost all of the studies, researchers have focused on the trends observed by changing variables while in this study not only is the same trends reported before witnessed, but the optimum value for each variable based on FOM is also proposed.

In the fifth stage, a new design for pin-fins (tapered pin-fin heat sinks) has been investigated and optimized. The schematic of a tapered pin-fin with its geometric parameters is indicated in Fig. 11 (a). In this regard, the diameter of fins is kept non-uniform and it is changed from the base to the tip. It is worth mentioning that all the fins have the same situation with their base diameter fixed and equal to the optimal value (d). Additionally other geometric parameters are in their optimal values ($N=26$, $\frac{t}{l} = 0.048$, $\frac{H}{l} = 0.0135$). The effect of employing tapered pin-fins on the hydrothermal performance of the pin-fin heat sink is shown in Fig. 11 (b). Considering Fig. 11 (b), both pressure drop and convective heat transfer coefficient increase as the tapered ratio (the ratio of the tip diameter to base diameter (r/R)) increases. By com-

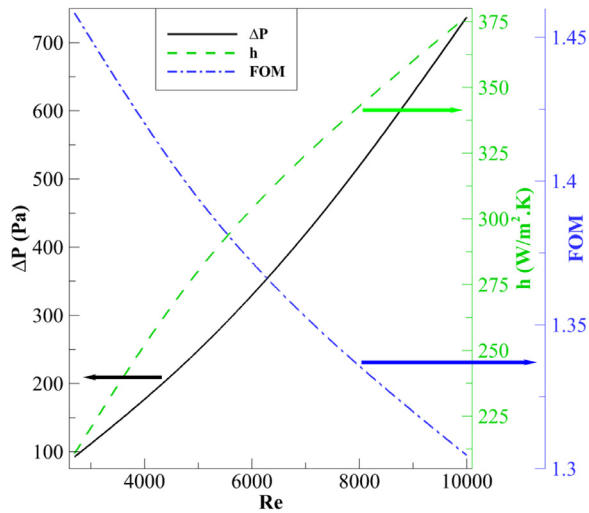


Fig. 13. The effect of Reynolds number on hydrothermal properties.

paring Fig. 11 to previous stages, it is obvious that the values of the convective heat transfer and pressure drop decrease by using

tapered pin-fin. It can also be seen that for $\frac{r}{R} = 0.3$, FOM rises up to 1.458, which is a relatively high value in comparison with previous stages, and the total performance of the system increases more than 17%. As a summary of all previous results and for a better understanding of the optimization stages, Fig. 12 is presented. It demonstrates a summary of changes in the pressure drop, convective heat transfer coefficient, and FOM of the optimum cases for each of the five stages studied so far. The parameters shown in Fig. 13 are defined as follows:

$$(\Delta P)_{opt}^* = \left(\frac{(\Delta P)_{opt_{phase i}} - (\Delta P)_{no-finned}}{(\Delta P)_{no-finned}} \right)^{1/3} \quad (17)$$

$$(h)_{opt}^* = \frac{(h)_{opt_{phase i}} - (h)_{no-finned}}{(h)_{no-finned}} \quad (18)$$

As can be seen, FOM increases step by step by moving forward from stage 1 to 5 so that the FOM value in the final step is 17.58% more than that in the first step. Nevertheless, both pressure drop and convective heat transfer coefficient reach a maximum value at stage 3, after which they start to decline. It is also obvious that the optimum convective heat transfer coefficient in stage 5 is less than that in stages 3 and 4. To deal with the problem, the effect of increasing Reynolds number on the hydrothermal performance of the optimum case in stage 5 has been studied. According

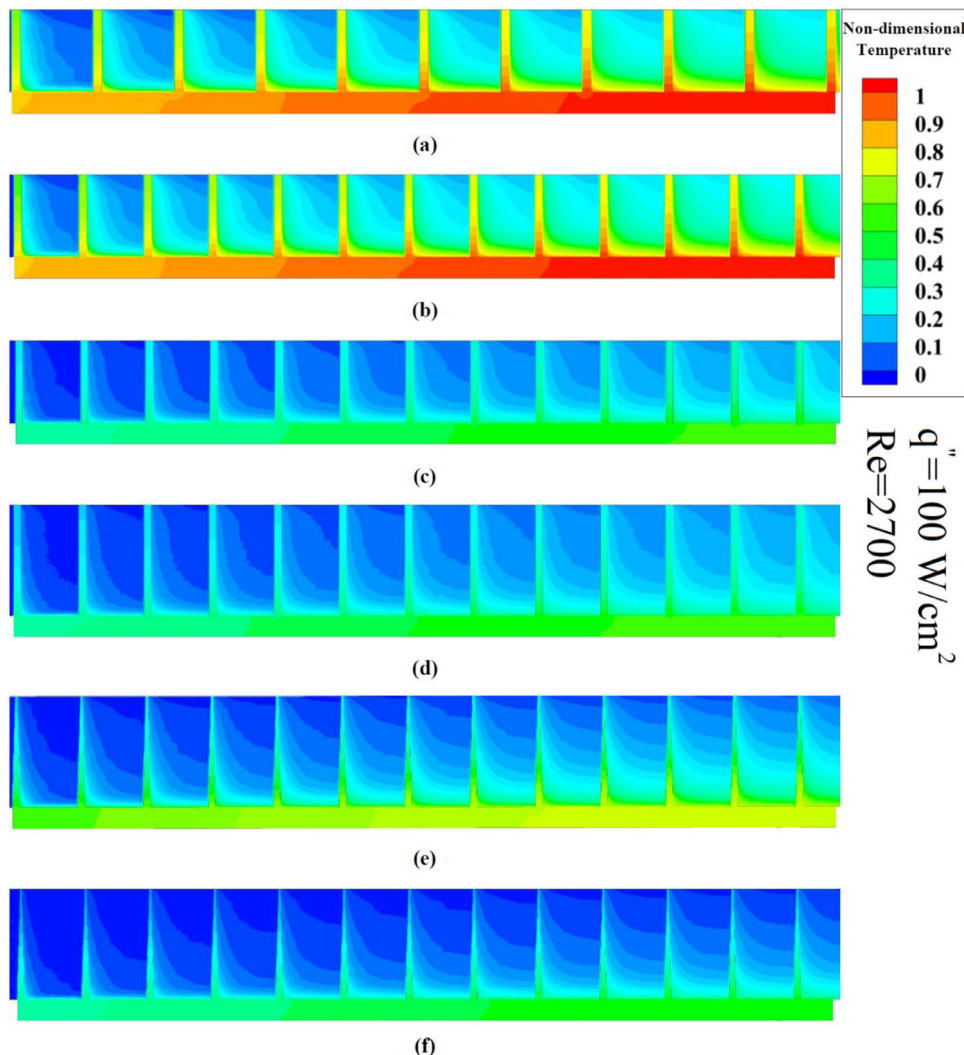


Fig. 14. Non-dimensional temperature contours of the optimum case for (a) the first stage, (b) the second stage, (c) the third stage, (d) the fourth stage, (e) the fifth stage, and (f) the fifth stage with the Reynolds number of 5000.

to Fig. 13, both pressure drop and convective heat transfer coefficient increase as Reynolds number increases. However, the increase in pressure drop is so sharp that causes a continuous decrease in FOM. Still, for $Re=5000$, both FOM (1.394) and convective heat transfer ($280.2 \frac{W}{m^2 K}$) coefficient are higher than the optimum values for stage 4. Therefore, it can be assumed that the optimal hydrothermal performance of the tapered pin-fin heat sink with the aforementioned Reynolds number is better than that of the simple, circular pin-fin heat sink.

To compare the thermal performance of the optimum cases in each phase, the contours of the non-dimensional temperature, $\frac{T - T_{min, global}}{T_{max, global} - T_{min, global}}$ are presented, from the side view in Fig. 14. According to Figs. 14 (a) and (b), due to relatively low values of the convective heat transfer coefficient of phases 1 and 2, the base temperature is rather high. Moreover, the interfacial temperature gradient in both the heat sink base and fins is relatively high. Given Figs. 14 (c) and (d), with improvements in the convective heat transfer coefficient for stages 3 and 4, the base temperature declines and the interfacial temperature distribution along the heat sink and fins becomes more uniform than that for previous cases. Nonetheless, considering Fig. 14 (e), an increase in the base temperature is observed, resulting from the decrease in the convective heat transfer coefficient. Finally, Fig. 14 (f) indicates that using tapered pin-fin heat sinks with a higher Reynolds number yields acceptable results when it comes to the heat sink interfacial temperature. Furthermore, this case exhibits the most uniform interfacial temperature distribution through the entire heat sink.

4. Conclusion

An inclusive optimization of pin-fin heat sinks was conducted in five stages in this study to find the best hydrothermal performance based on the FOM criterion. These geometric parameters are: the number of fins, the fin height, the fin diameter and the transverse pitch. Also, a new design of pin-fins (tapered pin-fins) has been investigated and its geometric parameter (tapered ratio) has been optimized. At the end of the study, the effect of increasing Reynolds number on the hydrothermal performance of the system has been investigated. The most prominent results of this paper have been listed below:

- Increasing the number and diameter of pin-fins increases both pressure drop and heat transfer coefficient, while increasing the fin height and transverse pitch results in a drop in both pressure drop and heat transfer coefficient.
- The values of the convective heat transfer and pressure drop decrease by using tapered pin-fins. Also, for $r/R = 0.3$, FOM rises up to 1.458, which is a relatively high value in comparison with previous stages
- The hydrothermal performance of a tapered pin-fin heat sink with ($N=26$, $d/l=0.0085$, $t/l=0.048$, $H/l=0.0135$ and $r/R=0.3$) has improved by 17.58% in comparison with conventional pin-fin heat sinks.
- Both pressure drop and convective heat transfer coefficient increase as Reynolds number increases. However, the increase in pressure drop is so sharp that causes a continuous decrease in FOM.

Declaration of Competing Interest

None.

CRediT authorship contribution statement

M. Ahmadian-Elmi: Supervision, Conceptualization, Methodology, Software, Writing – review & editing. **A. Mashayekhi:** Soft-

ware, Validation, Writing – original draft. **S.S. Nourazar:** Supervision, Writing – review & editing. **K. Vafai:** Supervision, Writing – review & editing.

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